NANDITA KUMARI

Website |Google Scholar | GitHub | Contact

EDUCATION

- **PhD in Geosciences**, Stony Brook University GPA **3.96**/4.0 [Fall'19-Present]
- **B. E in Geoinformatics**, College of Engineering Guindy, Anna University, Chennai, GPA **8.93**/10 [2018]

AWARDS

- **NASA Exploration Science Forum** Travel Award, 2023
- **Zonta International Amelia Earhart Fellowship,** 2023
- **Featured Early Career Presentation**, Lunar Exploration Analysis Group (LEAG) Meeting 2022
- **LEAG Travel Award**, 2022
- **Best Poster Award** in National Space Science Symposium 2019
- **Mike Barnsley Memorial Award**[1] by Alumni Association of CEG
- **Alumni Association Endowment Award 2007-2**[2] in Geoinformatics Engineering batch 2018
- **Merit Scholarship** from Army Welfare Association

INVITED TALK

VELAY MEETING, STONY BROOK UNIVERSITY,2023

PROFESSIONAL EXPERIENCES & SERVICES

CHAIR		*NASA EXPLORATION SCIENCE FORUM 2020, 2023*	
REVIEWER		*ICARUS, PSJ, GRL*	
EXTERNAL REVIEWER		*NASA PANEL 2023*	
EXECUTIVE SECRETARY		*NASA PANEL 2022*	
MENTOR		*GWISE*	[FALL'22-SPRING'23]
CHAIR		*LUNAR EXPLORATION ANALYSIS GROUP MEETING 2022*	
SUMMER INTERN		*LUNAR AND PLANETARY INSTITUTE*	[MAY'20-AUG'20]
VISITING STUDENT		*PHYSICAL RESEARCH LABORATORY*	[MAY'18-JUL'19]
SUMMER INTERN		*PHYSICAL RESEARCH LABORATORY*	[MAY'16-JUN'16]

PEER REVIEWED PUBLICATIONS

1. SURFACE CONDITIONS AND RESOURCE ACCESSIBILITY AT POTENTIAL ARTEMIS LANDING SITES 007 AND 011 (LINK)
2. GEOLOGICAL CHARACTERIZATION OF CHANDRAYAAN-2 LANDING SITE IN THE SOUTH POLE REGION OF THE MOON (LINK) [CO-AUTHORED]
3. THE SCIENTIFIC VALUE OF A SUSTAINED EXPLORATION PROGRAM AT THE ARISTARCHUS PLATEAU (LINK) [CO-AUTHORED]
4. TEMPERATURES OF THE LACUS MORTIS REGION OF THE MOON (LINK) [CO-AUTHORED]
5. DIFFERENT TRENDS OF VARIATION IN CHRISTIANSEN FEATURE ON THE LUNAR SURFACE : EFFECTS OF IRON CONTENT AND PARTICLE SIZE [IN REVIEW, JGR]
6. COMPOSITIONAL AND THERMOPHYSICAL PROPERTIES OF THE GRUITHUISEN REGION ON THE MOON [IN PREP]

[1] Annual award for highest marks in image processing in theory and lab

[2] Department rank 1 for the academic year 2016-2017

Non Peer Reviewed Publications

1. Space Weathering Across the Solar System: Lessons from the Moon and Outstanding Questions (Link) [Co-Authored]
2. Mission to Characterize Volatiles in Old, Cold, Permanently Shadowed Regions on the Moon (Link) [Co-Authored]

Conference Abstracts

1. Bulk Composition Characterization of The Lunar Surface Using Machine Learning: Applications to Ongoing and Upcoming Missions [NESF 2023]
2. Brewing Machine Learning Models For Accurate Analysis of Current and Future Datasets [LPSC 2023]
3. Procedure Standardization For Toolbox for Research and Exploration (TREX) Field Data Analysis [LPSC 2023]
4. Gruithuisen Domes: Silicic Volcanic Constructs on The Moon [LEAG 2022]
5. A Holistic View of Gruithuisen Domes – Silicic Constructs on the Moon [LPSC 2022]
6. Surface Illumination, Temperatures and Cold Traps at Two Potential Landing Sites Near the Lunar South Pole [NESF 2021]
7. Mid-Infrared Spectral Characteristics of Dust (Minerals and Meteorites) and Their Utility for Interpreting Remote-Sensing Data.[SOFIA Rock, dust and Ice Workshop 2021]
8. Episodic Volcanism and Fluvial Activities in a Floor-Fractured Crater at the Flanks of Arsia Mons [LPSC 2021]
9. Geological Diversity at Two Potential Landing Sites in the Lunar South Pole[LPSC 2021]
10. Artemis III EVA Opportunities on the Rim of de Gerlache Crater [NASA requested input for Artemis III Science Definition Team. Co-Authored]
11. Alternate Artemis III EVA Opportunities near de Gerlache Crater [NASA requested input for Artemis III Science Definition Team, Co-Authored]
12. Artemis III EVA Opportunities on the Lunar Farside near Shackleton Crater [NASA requested input for Artemis III Science Definition Team, Co-Authored]
13. Different Trends of Variation in Christiansen Feature on The Lunar Surface: Effects of Iron Content and Particle Size [NESF 2020]
14. Investigation of Variation in Christiansen Feature With Respect To Albedo on the Moon Using the Diviner, M^3 and Kaguya Datasets [LPSC 2020]
15. Global Distribution and Characteristics of Domical Craters on the Moon [LPSC 2019]
16. Complex Impact Craters within Lunar Basins and Their Implication [NSSS 2019]

Field Work

Kilbourne and Hunts Hole, SSERVI (RISE4) 2022;2023

San Francisco Volcanic Field, SSERVI (CLSE) 2022

Utah Yellowcat Site, SSERVI(TREX),2022

Mount Ascutny, *New England NEIGC Field Trip 2019*

Leadership Skills

Senator	| Graduate Student Organization of SBU	[2021-Present]
Student Director	| *CEG Tech Forum*	[2014-2018]
Director	| *Literary Club of Anna University*	[2018]
Core Member	| *Society for Geoinformatics Engineers*	[2014-2018]
Experience Manager	| *AIESEC International, Chennai*	[Jul'15– Jul'16]

Technical Skills

PROGRAMMING	Python(Data Analysis/Machine Learning, Proficient)
SOFTWARE	ArcGIS, QGIS, ENVI, ERDAS Imagine, ISIS, DaVinci
DATA PROFICIENCY	Earth: LANDSAT[3] (VIS, THERMAL, PAN), ASTER Thermal InfraRed Data; Moon: LRO (WAC, NAC, WAC-DTM, Diviner, LOLA), Chandrayaan-1(M^3), SELENE, KAGUYA ; Mars: MRO (CRISM, CTX, HiRISE), MGS(MOLA)

[3] LANDSAT Data from 5 to 8